Towards cylindrical cloak in nanosphere dispersed liquid crystal metamaterial

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ABSTRACT

We discuss the concept of infrared cloaking using nanosphere dispersed liquid crystal (NDLC) matematerial in cylindrical geometry. Preliminary results show that NDLC is a promising candidate for cloak design. Monte Carlo simulations are used for the design.

1. METAMATERIALS, CLOAKING AND TRANSFORMATION OPTICS

The concept of metamaterials was proposed in 1968 by Veselago[1] who has shown that Maxwell equations predict negative values of refractive index. However, this concept was not further explored because materials with such unusual properties can not be found in nature. The interest in the field of metamaterials started to grow steadily since the first experimental realization by Pendry[2] in 1998. The proposed metamaterial composed of thin metal wires and split ring resonator offered no possibility of tuning. On the contrary, the liquid crystal based metamaterial used by us offers a possibility of tuning. The reorientation of LC molecules causes the change of the permittivity of the host and influence the effective parameters of the metamaterial.

Since the first experimental realization of metamaterial many different approaches to obtain the negative effective refractive index were invented. Basing on them different types of metamaterial–based devices were designed.[3] One of the first applications for metamaterials was a perfect lens proposed by Pendry[4] that allows to image the object without aberrations and with sub–wavelength resolution.

New possibilities of tailoring the light flow became available with the use of transformation optics.[5] This method allows to mold the flow of light in practically arbitrary way using a particular spatial distributions of electromagnetic parameters of the medium. Using the form–invariance of Maxwell equations with respect to the space coordinate transformation one can shape the light trajectory and bend it in a similar way as it happens in the curved space in the general theory of relativity. To achieve this effect the spatial distribution of the permeability and permittivity tensor of the medium has to satisfy the conditions resulting from the coordinate transformation.

One of the applications of transformation optics is a long time dream of human civilization – invisibility. Novel metamaterials where the spatial distribution of optical parameters can be specifically defined allow to design/produce cloaking devices providing invisibility effect. The light flow can follow a desired trajectory so that the hidden object appears to be invisible. The omnidirectional spherical cloak was proposed.[6,7] However, in this case the conditions imposed on permeability and permittivity tensors are very hard to meet. Therefore the first experimental realization used a cylindrical cloak working only for one particular polarization of light. The design was based on the array of split–ring resonators and was operating in the microwaves range. The details of linear coordinate transformation and metamaterial can be found in Ref.[8] Similar devices operating in the visible range were proposed by Cai *et al.*[9] The same linear coordinate transformation was used but the metamaterial consisted of thin metal wires in a dielectric medium. The devices mentioned above were designed just for one polarization and cause a non–zero scattering due to impedance mismatch. To avoid this problem the higher order coordinate transformations can be used[10,11] that, on one hand result in a more complex distributions of the medium parameters but, on the other hand, cause less scattering.

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A different approach to cloaking is related to the concept of invisibility carpet proposed in paper.[12] It allows to "hide" the bump on the flat reflecting surface. Above the bump the layer of metamaterial with spatial distribution of optical parameters is placed that causes the light to reflect as from the flat surface. Big advantage of this kind of cloaking is that the properties of the material providing cloaking are not resonant. As a result the invisibility effect is achieved for the broad range of frequencies in contrast to the invisibility cloak where the proper values of material parameters occur close to the resonance and the operating range is very narrow.

The first experimental realization of the invisibility carpet was proposed in Ref.[13] The carpet was operating in millimeter wavelength region. More recent designs[14,15] allow for cloaking in near infra–red range and are based on silicon structures and photonic crystals.

The similarities between transformation optics and general relativity were pointed out by Leonhardt.[16] The link between the Fermat principle, stating that light chooses the trajectory with the shortest optical path, and the celestial mechanics in curved time–space was established by Genov.[17] It opens the way to investigate the orbital motion, strange attractors and chaos, in a controlled laboratory environment using the light. The optical–mechanical analogy enables direct studies of critical light/matter behavior around massive celestial bodies and, on the other hand, points toward the design of novel optical cavities and photon traps for application in microscopic devices and lasers systems. Basing on this principles the omnidirectional broadband photon absorber forming an optical black hole equivalent was designed.[18] Also, the light storing device was proposed where light can be accumulated and recovered from when needed.[19]

The effects mentioned above were designed using transformation optics where only the spatial coordinates undergo the transformation. There is also the possibility to involve time in the transformation. This allows us to hide not only the objects but also the events. By appropriate slowing down the light rays and speeding them up one can cause that some events can be undetectable for a distant observer.[20]

Besides those effects transformation optics can play more tricks. Is allows us to create the stereoscopic illusion so that an arbitrary object appears to be an object with completely different shape. Lai *et al.* was the first to numerically demonstrate this unusual optical phenomenon.[21] The principle lies in two distinct pieces of metamaterial: first one annihilates the light coming from the object and the second one recovers the light forming from it the image of another object chosen for illusion.

2. NANOSPHERE DISPERSED LIQUID CRYSTAL METAMATERIAL

The idea of the metamaterial that is used in this paper was started by Wheeler *et al.*[22] The material consisting of coated spheres was proposed as a medium with negative refractive index. The possibility of tuning of refractive index in this material was proposed by Khoo and co–workers.[23] The coated spheres were randomly dispersed in liquid crystal (LC) host. The reorientation of LC molecules allows to control the effective refractive index of nanodispersed liquid crystal (NDLC) metamaterial.

Scattering of the electromagnetic wave on a single coated sphere was analyzed using Mie scattering theory.[24] The scattered field in the dipole approximation was compared to the field produced by radiating dipole. From this comparison the electric and magnetic polarizabilities of a single coated sphere have been calculated. The effective parameters of an ensemble of coated spheres dispersed in NLC matrix were analyzed in the framework of effective medium theory. Using Maxwell Garnet mixing rule[25] that relates the properties of the spherical inclusions in the bulk material to the effective optical parameters the effective permittivity and permeability of NDLC metamaterial were retrieved. The resulting effective parameters are different from the parameters of the components forming the medium.

The proper choice of the parameters of the constitutive parameters is important for design of the effective material properties. The resulting effective refractive index of the NDLC media strongly depends on the properties of the building blocs: core and shell of the coated spheres and nematic liquid crystal (NLC) host.

The procedure described at the beginning of this paragraph yields in the following expressions for effective permittivity and permeability:[23]

$$\epsilon_{eff} = \epsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \tag{1}$$

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$$\mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}, \tag{2}$$

where N denotes number density of the spheres, $k_3 = \sqrt{\epsilon_3} k_0$, where ϵ_3 denotes permittivity of the liquid crystal host and $k_0 = 2\pi/\lambda_0$. λ_0 stands for the free space wavelength. According to the Mie theory in the dipolar (electric and magnetic) approximations the amplitudes of scattered electric and magnetic fields are proportional to coefficients a_1 and b_1, respectively. The detailed expressions can be found in Ref.[24] Those coefficients enter the expressions for effective permittivity and permeability of NDLC (1) and(2) and are responsible for the resonances.

The effective index of refraction was calculated as:

$$n_{eff} = \pm\sqrt{\epsilon_{eff}\ \mu_{eff}}, \tag{3}$$

where the root with a positive imaginary part should be chosen. Khoo et al. have shown[23] that for some values of the parameters of the model there is a range of frequencies ω and angles θ between directions of incident light wave vector and director of NLC for which

$$n'_{eff}(\omega, \theta) \equiv \Re\big(n_{eff}(\omega, \theta)\big) < 0. \tag{4}$$

The formulas given above are valid only for small number density $N = 3f/4\pi r_2^3$ of spheres (f is the filling factor) and in long–wavelength limit $2\pi/k_3 \gg r_2$.

To ensure the resonance in the effective permeability (2) we need to obtain high values of bulk permittivity of the material the core is made of.[22] Therefore we choose the core with radius r_1 to be made of polaritonic material (such as $LiTaO_3$) with relative bulk permittivity given by:

$$\epsilon_1(\omega) = \epsilon_1(\infty)\left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right), \tag{5}$$

where $\epsilon_1(\infty)$ denotes the high–frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively. The resonance in effective permittivity μ_{eff} occurs at relative bulk permittivity of the order of magnitude 10^2. Due to high value of bulk permittivity of the core material the resonance in the core is a bulk cavity resonance and is responsible for negative effective permeability of the assembly of spheres at particular electromagnetic wavelengths.

To obtain the resonant behavior of the effective permittivity (1) the shell with outer radius r_2 should have small negative value of the bulk permittivity. To this end we chose the material of the shell to be semiconductor with the relative bulk permittivity described by Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{6}$$

where ω_p denotes plasma frequency and γ_2 – the damping coefficient. Due to negative value of the permittivity for frequencies lower than the plasma frequency ω_p the resonance in the shell will be surface resonance and will not influence the core properties. The shell resonance is responsible for negative effective permittivity, at the frequency slightly smaller than ω_p. The negative effective refractive index is designed to occur at this frequency. The parameters of the materials creating the coated sphere and the sizes of the core and the shell are precisely chosen in such a way that they provide both negative effective permittivity and permeability in the same interval of frequencies.[22]

The tunability of effective refractive index is provided by changing the permittivity of the LC host. Permittivity of the liquid crystal host for extraordinary polarized wave depends on angle θ:[26]

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{7}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are permittivities for parallel and perpendicular orientations of director with respect to incident wave vector $\vec{k}_0$ in vacuum. In pure NLC host (without spheres) $k_3 = \sqrt{\epsilon_3}\, k_0$. The reorientation of the LC molecules and the resulting change of the angle θ cause a small change in the resonance conditions for effective permittivity and permeability and therefore influences the values of those parameters. Reorientation of NLC molecules can be induced by external electric or optical field or by anchoring forces.

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3. MONTE CARLO SIMULATIONS

Monte Carlo simulations[27,28] use random sampling to explore the multi–particle configuration space to provide statistical–mechanics description of the system. Our practical realization of those general ideas for LC system is described in Ref.[29] Interactions are described by the Lebwohl–Lasher effective Hamiltonian with Rapini-Papoular term. Potential energy U (divided by k_BT, where k_B – Boltzmann constant, T – absolute temperature) reads:

$$\frac{U}{k_BT} = -\frac{\xi}{k_BT} \sum_{<\vec{r},\vec{r}'>} P_2(\cos\beta(\vec{r},\,\vec{r}')) - \sum_{\vec{r}} \frac{E^2(\vec{r})}{k_BT} P_2(\cos\beta(\vec{r})) + \sum_{\vec{r}_W} \frac{\alpha(\vec{r}_W)}{k_BT} \sin^2\gamma(\vec{r}_W), \tag{8}$$

Here P_2 is the second–order Legendre polynomial. Parameter ξ is the coupling constant for NLC–NLC orientational interaction (in simulations we put $\xi = 25$), $E(\vec{r})$ is proportional to external electric field, $\alpha(\vec{r}_W)$ denotes the amplitude of anchoring forces acting on molecules located at the walls. $\beta(\vec{r},\,\vec{r}')$ denotes the relative angle between two molecules located at positions $\vec{r}$, $\vec{r}'$. $\beta(\vec{r})$ is the angle a molecule located at location $\vec{r}$ makes with the electric field direction. $\gamma(\vec{r}_W)$ denotes the angle between long axis of molecule at the wall and a fixed rubbing direction.

4. GRADIENT OF REFRACTIVE INDEX AND ITS TEMPERATURE DEPENDENCE

Large gradients of refractive index are of prime interest for design of molding of the flow of light. Modulated external electric field[29] and modulated anchoring forces[30] generate gradients of refractive index in NDLC systems and are important parameters for a design of molecular order required for an implementation of geometric conditions for guiding the flow of light in NDLC. In this section we consider the simplest case when inhomogeneity of anchoring forces is described by step function.[30] Regions of surface with strong anchoring ($\alpha = 50$) are placed close to regions without anchoring ($\alpha = 0$):

$$\alpha(x) = \begin{cases} 50 & x \le 50 \\ 0 & x > 50 \end{cases} .$$

In Ref.[30] we have shown that NDLC system has much larger gradients of effective refractive index $|\partial\langle n'_{eff}\rangle/\partial x|$ then pure NLC. Below we investigate temperature dependence of this gradient. Critical temperature was calculated based on order parameter using Monte Carlo simulations without anchoring: $k_BT_c/\xi = 1.15$ (more details can be found in Ref.[31]). In Fig. 1 2D maps of local refractive index for $\lambda = 2875$ nm at different temperatures are presented. Fig. 2 shows the profiles $\langle n'_{eff}\rangle(x)$ calculated along the lines of Ref.[30] based on 2D maps from Fig. 1.

Presented preliminary results indicate a strong temperature dependence of optical properties of NDLC system with modulated anchoring forces. In Fig. 3 gradients $\partial\langle n'_{eff}\rangle/\partial x$ for $T = 0.36\ T_c$ and $T = 0.71\ T_c$ are shown. At the lower temperature the gradient is much larger than at the higher temperature.

5. TOWARDS THE DESIGN OF CLOAKING IN NDLC FOR CHOSEN INFRARED WAVELENGTHS

We discuss some aspects of the design of an infrared cylindrical cloak using NDLC metamaterial. Devices based on liquid crystal potentially offer the possibility of real–time control of invisibility.

Consider a cylindrical NLC cell with inner and outer radii a and b, filled with NDLC metamaterial (Fig. 4). The optical transformation for cylindrical systems and general requirements for tensor components of permittivity and permeability which satisfy cloaking condition were discussed in Refs.[3,9] For transverse magnetic (TM) polarization of incident light those conditions read:

$$(\mu'_{eff})_z(\varepsilon'_{eff})_r = \left(\frac{b}{b-a}\right)^2\left(\frac{r-a}{r}\right)^2, \tag{9}$$

$$(\mu'_{eff})_z(\varepsilon'_{eff})_\Theta = \left(\frac{b}{b-a}\right)^2. \tag{10}$$

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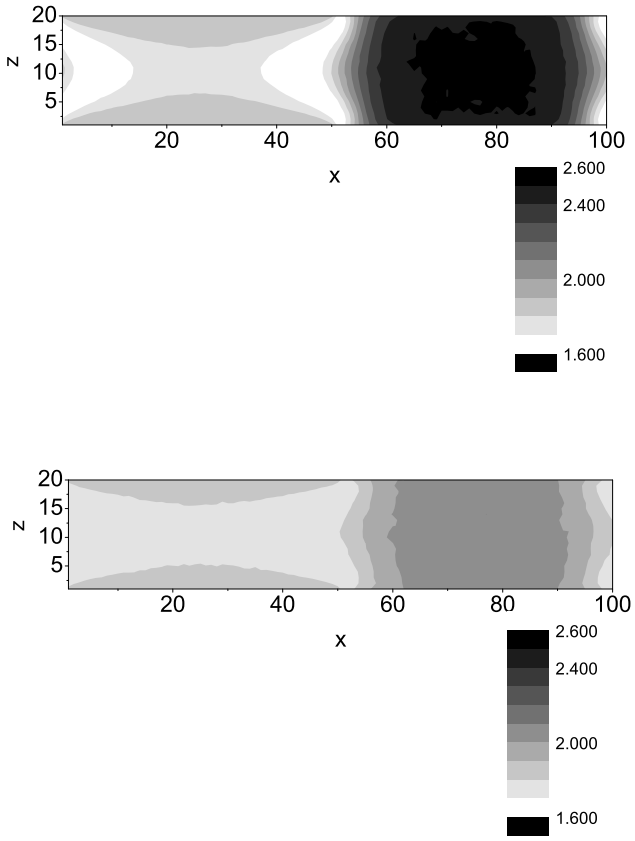

Figure 1. 2D maps of $n'_{eff}(x,z)$. $\lambda = 2875$ nm, $E \approx E_F$, $T = 0.36\,T_c$ (top) and $T = 0.71\,T_c$ (bottom).

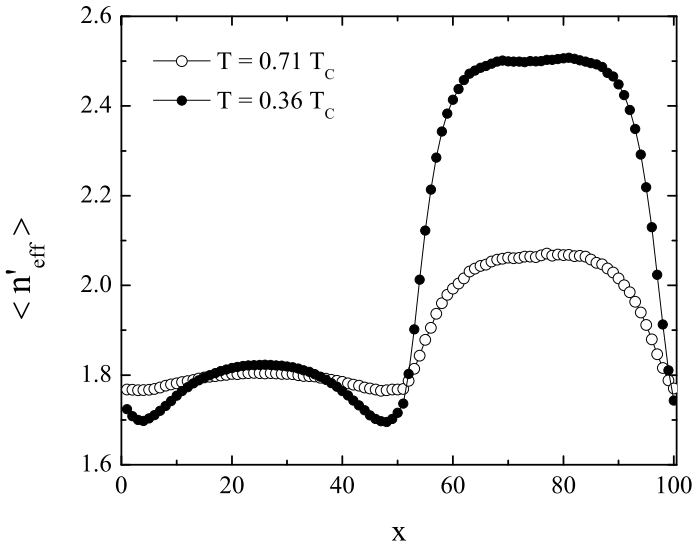

Figure 2. Profiles of averaged refractive index $\langle n'_{eff}\rangle(x)$ for $\lambda = 2875$ nm, $E \approx E_F$, for $T = 0.36\,T_c$ and $T = 0.71\,T_c$.

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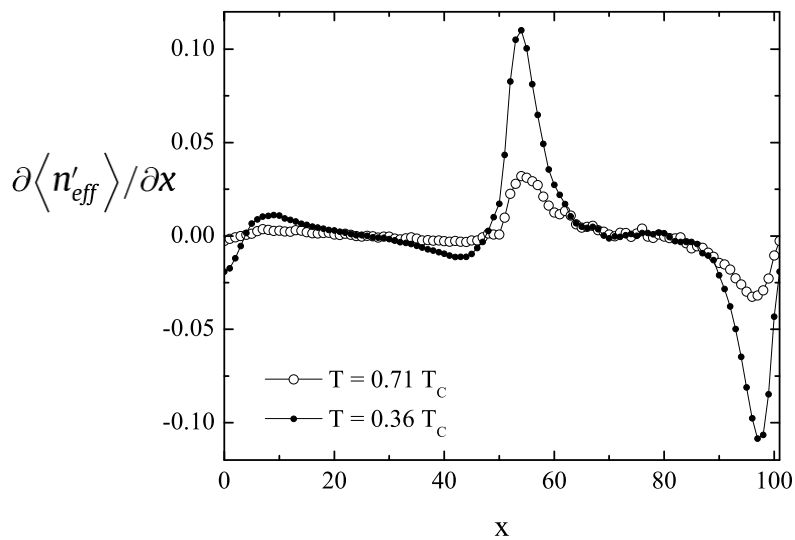

Figure 3. Plot of $\partial\langle n'_{eff}\rangle/\partial x$ for NDLC with modulated anchoring, $\lambda = 2875$ nm, $E \approx E_F$, for $T = 0.36\ T_c$ and $T = 0.71\ T_c$.

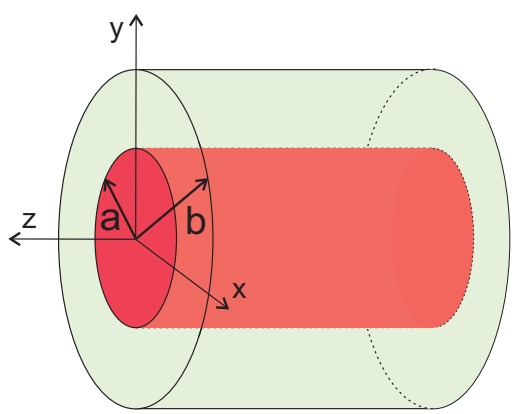

Figure 4. Cylindrical NLC cell with inner and outer radii a and b, filled with NDLC metamaterial.

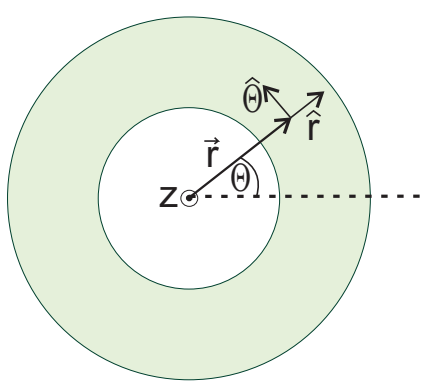

Figure 5. Cylindrical coordinates.

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Indices z, r, Θ refer to cylindrical coordinates (Fig. 5).

If the long axes of NLC molecules are in $x - y$ plane then the wave with TM polarization incidents as an extraordinary wave onto the NLC host. Based on local orientation of NLC molecules local tensor of permittivity of the host ϵ_3 is calculated:

$$\varepsilon_3 = \begin{pmatrix} \varepsilon_\perp + \Delta\varepsilon \sin^2\theta & \Delta\varepsilon \sin\theta\cos\theta & 0 \\ \Delta\varepsilon \sin\theta\cos\theta & \varepsilon_\perp + \Delta\varepsilon\cos^2\theta & 0 \\ 0 & 0 & \varepsilon_\perp \end{pmatrix}. \tag{11}$$

In the next step effective parameters ϵ_{eff}, μ_{eff} are calculated according to eq. (1, 2).

The design consists in finding a set of parameters (size of nanospheres, ordinary and extraordinary refractive indices of NLC host, filling factor, size of cloak a and b, configuration of NLC particles, wavelength) for which the cloaking requirements can be met. Preliminary studies show that such a set of parameters can be found. This topic and the study of thermal fluctuations and their impact on cloaking effect will be the subject of a separate paper.

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